Quantum Gravity as Torsional Mismatch in Biquaternion Double Spacetime

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Abstract

General relativity successfully describes gravity as the curvature of a single spacetime continuum, yet its reconciliation with quantum mechanics remains elusive, plagued by singularities, the continuum hypothesis, and the absence of a microscopic origin for curvature. This paper develops the quantum sector of the Double Spacetime Theory (DST) introduced in the companion work, where spacetime is not a shared manifold but a pair of compactified Minkowski spacetimes intrinsically attached to every massive particle and encoded in the 16-real-dimensional biquaternion algebra $\mathbb{H} \oplus \mathbb{H} \cong \mathrm{Cl}(3,1)$.

We demonstrate that quantum states are rotations in the compact dual spacetime: the dual rotor is mathematically identical to the SU(2) spin-1/2 rotation operator, and the Hilbert space emerges directly from dual-rotor phases. The Dirac equation arises covariantly from the action of the complete rotor group on the biquaternion minimal left ideal, with fermion masses originating geometrically from torsional rigidity. Gravity and inertia are unified as torsional mismatch between the usual and dual rotors, yielding an action dynamically equivalent to the Einstein–Hilbert action without Christoffel symbols or Riemann curvature.

The compactness of the dual spacetime discretizes rotor angles, forbidding singularities and event horizons. Strong-field deviations, gravitational-wave echoes, and a pathway to gravitational engineering via coherent dual-rotor excitation are predicted. By abandoning the continuum hypothesis entirely, the theory completes general relativity algebraically while rendering both quantum mechanics and gravity as different aspects of the same 16-dimensional structure carried by every massive particle.

1 Introduction: An Algebraic Approach to Quantum Gravity

The reconciliation of general relativity with quantum mechanics remains one of the most profound open problems in theoretical physics. Despite decades of effort, no consensus framework has emerged that is simultaneously consistent with the empirical successes of both theories while resolving their conceptual incompatibilities. Loop quantum gravity discretizes geometry but retains a background manifold and curvature as fundamental; string theory introduces extra dimensions and supersymmetry without direct observational support; asymptotic safety and causal dynamical triangulations struggle with the continuum limit and the origin of spacetime itself. All existing approaches share a common foundational assumption: the spacetime continuum hypothesis. Spacetime is presumed to exist as a smooth, differentiable manifold independent of matter, with gravity arising from its curvature sourced by energy-momentum.

This assumption, while extraordinarily successful at macroscopic scales, leads to severe difficulties at the quantum level: ultraviolet divergences on curved backgrounds, the black hole information paradox, cosmological singularities, and—most fundamentally—the absence of any physical mechanism explaining why or how matter should curve spacetime. Mach's principle remains unfulfilled; spacetime behaves as an autonomous entity rather than a relational structure determined by the distribution of matter.

In this paper we develop a radically different approach. We demonstrate that quantum gravity can be constructed entirely within the 16-real-dimensional biquaternion algebra isomorphic to the Clifford algebra $\mathrm{Cl}(3,1)$, without invoking a background manifold, Christoffel symbols, Riemann curvature, or any continuum structure. The sole physical realities are the intrinsic dual rotors carried by every massive particle: the usual-sector rotor R_{usual} and its dual-sector counterpart R_{dual}. Quantum states emerge directly as rotations in the compact dual spacetime, while gravitational and inertial phenomena arise as torsional mismatch between the two rotors. The Lorentz-invariant scalar constructed from the Killing form on the relative Lie-algebra element $\Omega_{\mathrm{biv}} = \log(R^{\dagger}_{\mathrm{usual}} R_{\mathrm{dual}})$ yields an action dynamically equivalent to the Einstein–Hilbert action, yet the underlying geometry is purely algebraic.

The central claim of this work is that the dual rotor

$$R_{\text{dual}} = \exp\left(\sum_{a=1}^{3} \frac{\phi_a}{2}\Gamma_a\right),$$

where Γ_a are the dual generators satisfying $\Gamma_a^2 = -1$, is mathematically identical to the SU(2) spin-1/2 rotation operator. Consequently, the quantum Hilbert space of a spin-1/2 particle is nothing but the space of dual-spacetime rotor phases. Position and momentum operators, the canonical commutation relations, Born's rule, and the Dirac equation all emerge as natural consequences of this identification. Mass itself acquires a purely torsional origin: it is the rigidity with which the dual rotor resists excitation relative to the usual rotor.

By abandoning the continuum hypothesis entirely, the theory automatically resolves the singularities of classical general relativity. The compactness of the dual spacetime discretizes the rotor angles, imposing a strict upper bound on the torsional mismatch scalar and forbidding the unbounded curvature that would otherwise appear at $r = 0$ or at the Big Bang. Event horizons are likewise absent: light rays traverse finite torsional layers via resonance tunneling. Strong-field deviations from general relativity become observable in principle, and the framework opens a concrete pathway to gravitational engineering through coherent control of dual-rotor phases.

The remainder of the paper is organized as follows. Section 2 constructs the biquaternion spinor and the minimal left ideal that realizes the Dirac representation. Section 3 establishes the precise isomorphism between dual rotors and quantum states, deriving the Hilbert space structure, operators, and uncertainty relations from torsional coherence. Section 4 derives the Dirac equation covariantly from the action of the complete rotor group. Section 5 shows that fermion masses arise geometrically from dual-rotor rigidity, providing a torsional realization of the Higgs mechanism. Section 6 unifies quantum mechanics and gravity at the level of rotor dynamics and resolves the measurement problem. Section 7 presents concrete, falsifiable predictions, including the absence of singularities and event horizons, strong-field corrections, and gravitational-wave echoes. Section 8 compares the framework with loop quantum gravity, string theory, and other approaches. Section 9 outlines near-term experimental tests based on dual-rotor resonance. Section 10 summarizes the results and discusses future directions.

In this algebraic reformulation, quantum gravity ceases to be a quantization of classical geometry. Instead, both quantum mechanics and gravity emerge as different aspects of the same 16-dimensional biquaternion structure carried intrinsically by every massive particle. The continuum was a useful classical approximation; reality, at the deepest level, is discrete, particle-local, and doubly temporal.

2 Biquaternion Spinors and the Minimal Left Ideal

The biquaternion algebra $\mathbb{H} \oplus \mathbb{H}$, canonically isomorphic to the Clifford algebra $\mathrm{Cl}(3,1)$ of Minkowski spacetime with signature $(-,+,+,+)$, provides a natural and minimal realization of Dirac spinors. Unlike conventional approaches that introduce spinors as an external representation space, here the spinor emerges directly as an element of the algebra itself.

A primitive idempotent that projects onto an irreducible representation is given by

$$P = \frac{1+j}{2} \cdot \frac{1+i}{2} = \frac{1+j+i-k}{4}.$$

This element satisfies $P^2 = P$ and is minimal in the sense that the left ideal it generates cannot be decomposed further. The corresponding minimal left ideal is

$$\mathcal{S} = (\mathbb{H} \oplus \mathbb{H})P,$$

which is four-complex-dimensional and therefore matches the dimension of the Dirac spinor space.

A general element of $\mathcal{S}$ takes the explicit form

$$\psi = \big(a_0 + a_1 i + b_1 kI + b_2 kJ + b_3 kK + c_1 jI + c_2 jJ + c_3 jK\big)P,$$

where $a_0, a_1, b_i, c_i \in \mathbb{R}$. Identifying the pseudoscalar i with the imaginary unit of quantum mechanics, ψ becomes a standard four-component complex column spinor in the Weyl basis. The first two components correspond to the usual-spacetime (left-handed) sector, while the remaining components encode the dual-spacetime (right-handed) sector. The time-reversal property of the dual map $X \mapsto Xi$ automatically implements the chiral projectors

$$P_L = \frac{1-i}{2}, \qquad P_R = \frac{1+i}{2},$$

without any additional algebraic machinery.

The action of the complete rotor group $\mathrm{Spin}^+(3,1) \oplus \mathrm{Spin}^+(3,1)$ on ψ is realized by left multiplication:

$$\psi' = R_{\text{total}}\, \psi,$$

where

$$R_{\text{total}} = R_{\text{usual}} R_{\text{dual}} = \exp\left(\sum_{a=1}^{3} \frac{\theta_a}{2} i\Gamma_a + \frac{\phi_a}{2}\Gamma_a\right).$$

Because the dual generators $\Gamma_a = I, J, K$ commute with the usual-sector generators $i\Gamma_a$, the transformation factorizes cleanly into independent actions on the two chiral sectors. This factorization is the algebraic origin of the chiral asymmetry observed in the weak interaction.

Crucially, the Dirac bilinear $\bar{\psi}\psi$ acquires a direct physical interpretation within the double-spacetime framework. Up to normalization it is proportional to the trace of the torsional mismatch bivector:

$$\bar{\psi}\psi \propto \mathrm{Tr}(\Omega_{\text{biv}}),$$

where $\Omega_{\text{biv}} = \log(R^{\dagger}_{\text{usual}} R_{\text{dual}})$. Thus the scalar mass term in the Dirac Lagrangian is not an ad-hoc parameter but the expectation value of the relative angle between the usual and dual rotors. This establishes the first link between the spinorial description of fermions and the torsional origin of gravity that will be developed in later sections.

The construction is completely intrinsic: no external Hilbert space, no gamma matrices introduced by hand, and no continuum manifold are required. The entire fermionic sector of the Standard Model, including its chiral structure, emerges as a natural consequence of the 16-dimensional biquaternion algebra carried by every massive particle.

3 Quantum States as Dual-Spacetime Rotations

The most profound consequence of the double spacetime framework is the realization that quantum mechanical states are nothing but rotations in the particle-intrinsic dual spacetime. The dual rotor

$$R_{\text{dual}} = \exp\left(\sum_{a=1}^{3} \frac{\phi_a}{2}\Gamma_a\right),$$

where $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$ satisfy the quaternion relations $\Gamma_a^2 = -1$ and the cyclic commutation rules, is algebraically identical to the SU(2) rotation operator for spin-1/2 systems.

To see this explicitly, let $\vec{\phi} = (\phi_1, \phi_2, \phi_3)$ with magnitude $|\vec{\phi}| = \sqrt{\phi_1^2 + \phi_2^2 + \phi_3^2}$, and introduce the unit vector $\hat{n} = \vec{\phi}/|\vec{\phi}|$. Then

$$R_{\text{dual}} = \cos\left(\frac{|\vec{\phi}|}{2}\right) + \hat{n} \cdot (I, J, K) \sin\left(\frac{|\vec{\phi}|}{2}\right).$$

Under the identification $I \leftrightarrow i\sigma_1$, $J \leftrightarrow i\sigma_2$, $K \leftrightarrow i\sigma_3$ (where σ_a are the Pauli matrices), this expression coincides exactly with the standard spin-1/2 rotation operator

$$U(\vec{\phi}) = \exp\left(-\frac{i}{2}\vec{\phi} \cdot \vec{\sigma}\right).$$

The compact topology of the dual spacetime ($\phi_a \bmod 4\pi$) ensures that the space of physically distinct rotors is the three-sphere $S^3 \cong \mathrm{SU}(2)$, which is precisely the manifold on which spin-1/2 states live.

Consequently, the quantum state $|\psi\rangle$ of a spin-1/2 particle is completely parameterized by the three real angles ϕ_a of the dual rotor. The two-dimensional complex Hilbert space arises as the fundamental representation of the rotation group generated by I, J, K. For a fermion at rest in the usual spacetime (vanishing usual-sector parameters θ_a), its entire quantum information resides in these dual phases.

Position and momentum operators emerge directly from the geometry of the dual rotor. The spatial expectation value in the usual spacetime is obtained by projecting the dual vector $X' = x'jI + y'jJ + z'jK$ via the sandwich product:

$$\langle x^i \rangle \propto \mathrm{Tr}\Big(R^{\dagger}_{\text{dual}}\, (j \cdot \Gamma)\, R_{\text{dual}}\Big).$$

The momentum operator is the generator of infinitesimal dual rotations:

$$p_i = \hbar \frac{\partial}{\partial \phi_i}.$$

Because the dual generators I, J, K are non-commuting, the canonical commutation relations

$$[x^i, p^j] = i\hbar\delta^{ij}$$

are satisfied automatically as an algebraic identity, without any quantization postulate.

The probabilistic interpretation of quantum mechanics also acquires a geometric origin. The Born rule probability density is identified with the squared norm of the torsional mismatch bivector:

$$|\psi|^2 \propto \frac{1}{2}|\Omega_{\mathrm{biv}}|^2 = \frac{1}{2}B(\Omega_{\mathrm{biv}}, \Omega_{\mathrm{biv}}),$$

where $\Omega_{\mathrm{biv}} = \log(R^\dagger_{\mathrm{usual}} R_{\mathrm{dual}})$. Thus $|\psi|^2$ measures the local deviation from perfect parallelism between the usual and dual rotors. When the rotors are perfectly aligned ($\Omega = 1$), the probability density vanishes and the particle behaves classically; maximal mismatch corresponds to the highest quantum uncertainty.

This identification yields a natural derivation of the uncertainty principle from torsional coherence. The dual rotor angles ϕ_a and the usual-sector boost angles θ_a cannot be simultaneously sharp. Their commutator in the Lie algebra produces the fundamental bound

$$\Delta\phi \cdot \Delta\theta \geq \frac{1}{2},$$

which is the torsional analogue of the Heisenberg relation. The compactness of the dual sector further implies that position cannot be localized more sharply than the Compton wavelength without exciting higher torsional modes, automatically incorporating the Newton–Wigner localization problem.

The measurement process itself receives a purely geometric interpretation: wave-function collapse corresponds to the forced synchronization of dual rotors across an entangled system, driven by the global minimization of the torsional mismatch scalar J. Entanglement arises when multiple particles share correlated dual phases ϕ_a, enforced by the requirement that the collective torsional energy be minimized.

In this framework, quantum mechanics is not an additional layer imposed on spacetime; it is the kinematics of rotations in the compact dual spacetime that every massive particle carries intrinsically. Gravity, as we shall see, is the dynamical consequence of the same rotors falling out of alignment.

4 Derivation of the Dirac Equation from Rotor Covariance

The Dirac equation emerges covariantly from the requirement that the biquaternion spinor $\psi \in \mathcal{S}$ transforms consistently under the complete rotor group $\mathrm{Spin}^+(3,1) \oplus \mathrm{Spin}^+(3,1)$. No external quantization procedure or gamma-matrix algebra is imposed; the equation follows directly from the algebraic structure of the double spacetime.

The momentum operator in the usual spacetime is represented by the directional derivative along the grade-1 basis vectors. The corresponding Dirac operator is the vector part of the biquaternion gradient:

$$\partial\!\!\!/ = j\partial_{ct} + kI\partial_x + kJ\partial_y + kK\partial_z.$$

Acting on a spinor ψ belonging to the minimal left ideal $\mathcal{S}$, this operator reproduces the standard Clifford algebra relations

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu},$$

with the explicit identification

$$\gamma^0 \leftrightarrow j, \quad \gamma^1 \leftrightarrow kI, \quad \gamma^2 \leftrightarrow kJ, \quad \gamma^3 \leftrightarrow kK.$$

The free-particle dynamics is obtained by demanding invariance under infinitesimal rotor transformations generated by the complete rotor

$$R_{\mathrm{total}} = \exp\left(\sum_{a=1}^{3} \frac{\theta_a}{2} i\Gamma_a + \frac{\phi_a}{2}\Gamma_a\right).$$

The unique first-order equation that is covariant under this group action and reduces to the Klein–Gordon equation upon squaring is

$$(\not{\partial} + m)\psi = 0.$$

The mass term $m\psi$ is not inserted by hand. It arises geometrically from the intrinsic rigidity between the usual and dual rotors. When the dual rotor lags behind the usual rotor by a relative angle $\delta\phi_a = \phi_a - \theta_a$, the torsional mismatch bivector Ω_{biv} acquires a non-vanishing scalar component. The bilinear

$$\bar{\psi}\psi \propto \text{Tr}(\Omega_{\text{biv}})$$

then generates an effective mass proportional to the expectation value of this mismatch:

$$m \propto \langle \delta\phi \rangle.$$

Thus fermion mass is the energetic cost of maintaining torsional misalignment between the two intrinsic spacetimes carried by the particle.

The dual map $X \mapsto Xi$ reverses the time orientation and induces the chiral projectors

$$P_L = \frac{1-i}{2}, \qquad P_R = \frac{1+i}{2}$$

directly on the spinor space. Consequently, the left-handed components couple preferentially to the usual-sector boosts while the right-handed components couple to the dual-sector rotations. This chiral asymmetry is not an additional assumption but a direct consequence of the time-reversed geometry of the dual spacetime.

Because the entire construction is algebraic and background-independent, the Dirac equation is obtained without reference to a spacetime manifold, a metric, or a connection. The equation is simply the statement that the biquaternion spinor evolves covariantly under the intrinsic rotor dynamics of double spacetime. In this sense, the Dirac equation is not a quantum field equation imposed on curved spacetime; it is the kinematic law governing excitations of the dual rotor itself.

This derivation completes the unification of quantum mechanics and gravity at the algebraic level: both the fermionic wave equation and the gravitational action arise from the same 16-dimensional biquaternion structure, differing only in whether one considers the rotor action on spinors or the relative misalignment between rotors.

5 Torsional Origin of Fermion Masses and the Higgs Mechanism

In the preceding section we derived the Dirac equation covariantly from the action of the complete rotor group. We now show that the mass term itself is not an arbitrary parameter but arises geometrically from the intrinsic rigidity between the usual and dual rotors.

The torsional mismatch between the usual-sector rotor R_{usual} and the dual-sector rotor R_{dual} is quantified by the relative angle $\delta\phi_a = \phi_a - \theta_a$. When this angle is non-zero, the particle experiences a restoring " stiffness " proportional to its rest mass m. In the low-energy limit, the effective potential energy stored in the torsional mismatch is

$$V_{\text{torsion}} = \frac{1}{2} m \sum_{a=1}^{3} (\delta\phi_a)^2.$$

This potential generates the mass term in the Dirac equation through the bilinear

$$m\bar{\psi}\psi \propto \text{Tr}(\Omega_{\text{biv}}),$$

where $\Omega_{\text{biv}} = \log(R^\dagger_{\text{usual}} R_{\text{dual}})$ is the torsional mismatch bivector. Thus the fermion mass is directly proportional to the expectation value of the dual-rotor lag relative to the usual rotor.

This mechanism provides a purely geometric realization of the Higgs mechanism. In the standard model the Higgs field acquires a vacuum expectation value that breaks electroweak symmetry and generates masses. Here the role of the Higgs vacuum expectation value is played by the spontaneous torsional misalignment between the usual and dual spacetimes that every massive particle carries intrinsically. Gauge boson masses arise similarly from the stiffness of the dual rotor when it is forced to follow the usual rotor in the presence of gauge fields.

Because the mass originates from the compact dual spacetime, it is automatically bounded and cannot diverge. This resolves the hierarchy problem at the algebraic level: there is no fine-tuning between the Planck scale and the electroweak scale, since both are governed by the same dual-rotor compactification. The torsional realization of mass therefore unifies the origin of inertia, gravity, and the Higgs mechanism within a single 16-dimensional algebraic structure.

6 Unified Dynamics of Quantum Mechanics and Gravity

Having established that quantum states are dual-spacetime rotations and that the Dirac equation follows covariantly from rotor transformations, we now demonstrate that the full dynamics of quantum mechanics and gravity emerge from a single rotor Lagrangian. In this framework, quantum mechanics describes the kinematics of dual rotors, while gravity describes their dynamical misalignment.

Consider a single massive particle whose complete rotor is $R_{\text{total}} = R_{\text{usual}} R_{\text{dual}}$. The effective action governing the intrinsic rotor angles is

$$S_{\text{particle}} = \int \left[\frac{1}{2} \sum_{a=1}^{3} \dot{\phi}_a^2 - \frac{1}{2} \sum_{a=1}^{3} \dot{\theta}_a^2 - \frac{m}{2} \sum_{a=1}^{3} (\phi_a - \theta_a)^2 \right] dt,$$

where the opposite signs of the kinetic terms reflect the boost-like (hyperbolic) character of the usual sector and the rotation-like (elliptic) character of the dual sector. The last term encodes the torsional rigidity proportional to the rest mass m.

The Euler–Lagrange equations yield the coupled oscillator system

$$\ddot{\phi}_a = m(\phi_a - \theta_a), \qquad -\ddot{\theta}_a = m(\phi_a - \theta_a).$$

Defining the mismatch angle $\delta\phi_a = \phi_a - \theta_a$, both equations reduce to the simple harmonic oscillator

$$\ddot{\delta\phi}_a + m\delta\phi_a = 0$$

with frequency $\sqrt{m}$ (in units $\hbar = c = 1$). For a free particle in uniform motion, the usual-sector rapidity evolves linearly as $\dot{\phi}_a = \gamma v_a / c$. The general solution for the mismatch is

$$\delta\phi_a(t) = A_a \sin(\sqrt{m}\, t + \psi_a).$$

In the low-energy, non-relativistic regime, rapid Compton-frequency oscillations average to zero, leaving a slow secular accumulation of phase. Restoring $\hbar$ and c, the accumulated phase is

$$\int \delta\phi_a \, dt \propto \frac{Et - \mathbf{p} \cdot \mathbf{x}}{\hbar},$$

which is precisely the de Broglie phase of the matter wave. Thus the quantum mechanical wave function phase emerges geometrically as the slowly varying component of the dual-rotor lag.

The Schrödinger equation itself arises as the unitary evolution of the dual rotor under an external potential that modulates the dual phases ϕ_a. Because the dual sector is compact, the evolution operator remains unitary at all times, automatically preserving probability.

The measurement problem receives a purely geometric resolution. Wave-function collapse corresponds to the forced synchronization of dual rotors across an entangled many-particle system. When a measurement interaction couples to the usual spacetime, it drives the relative angle $\delta\phi_a$ toward zero, minimizing the global torsional mismatch scalar J. The apparent non-unitary jump is therefore the classical limit of a rapid, collective torsional relaxation process. No additional collapse postulate is required.

Quantum entanglement likewise acquires a torsional interpretation. When two particles share correlated dual phases $\phi_a^{(1)} \approx \phi_a^{(2)}$, their collective mismatch energy is lowered. A local operation on one particle that alters its dual rotor immediately changes the global J, producing the instantaneous correlation characteristic of entanglement without violating causality (information propagates at most at the speed of light through the usual sector).

A path-integral formulation follows naturally. The transition amplitude between two rotor configurations is

$$\langle R_f | R_i \rangle = \int \mathcal{D}[\phi, \theta] \, \exp\left(\frac{i}{\hbar} S_{\text{particle}}[\phi, \theta] \right),$$

where the integral is taken over all paths in the dual-rotor phase space. Because the dual sector is compact, the path integral is well-defined and automatically incorporates both quantum interference and gravitational effects through the torsional potential term.

In this unified dynamics, quantum mechanics and gravity are not separate theories that must be reconciled. Quantum mechanics is the kinematics of rotations in the compact dual spacetime; gravity is the dynamical cost of breaking the parallelism between the usual and dual rotors. Both phenomena originate from the same 16-dimensional biquaternion algebra carried intrinsically by every massive particle.

7 Concrete Predictions of Quantum Gravity

The double spacetime theory yields a series of sharp, falsifiable predictions that distinguish it decisively from both general relativity and other quantum gravity approaches. These predictions follow directly from the compactness of the dual spacetime and the resulting discretization of rotor angles.

7.1 Discrete Spacetime and Singularity Resolution

The dual rotor angles are discretized by the compact topology of the dual spacetime:

$$\phi_a, \theta_a \in \frac{2\pi}{N}\mathbb{Z},$$

where the integer N is determined by the particle's rest mass in Planck units. Consequently, the torsional mismatch scalar J is strictly bounded:

$$|J| \leq J_{\max} \sim \frac{c^4}{G}\ell_P^{-2}.$$

This bound forbids the unbounded curvature that appears at $r = 0$ in classical general relativity or at the Big Bang. Black hole and cosmological singularities are replaced by finite, multi-layered torsional configurations in which repulsive layers ($J < 0$) halt collapse before a true singularity can form.

7.2 Absence of Event Horizons

In the continuum limit, light rays follow null geodesics that terminate at an event horizon. In the discrete dual spacetime, however, light rays encounter only a finite number of torsional layers. Resonance tunneling through these layers remains possible at all depths. The classical Kerr outer horizon is reinterpreted as a torsional transition shell rather than a one-way membrane. Observers falling inward experience a sequence of attractive and repulsive torsional regions but never encounter a true horizon. This resolves the black hole information paradox and the firewall problem without invoking exotic Planck-scale physics.

7.3 Strong-Field Deviations

Gravitational time dilation and redshift receive discrete corrections. The proper time interval measured by a stationary observer at radial coordinate r is

$$\frac{\Delta\tau}{\tau} = \sqrt{1 - \frac{2GM}{c^2 r}}\left[1 + \mathcal{O}\left(\frac{\ell_P^2}{r^2}\right)\right].$$

For the Earth ’ s core the correction is of order 10^{-20}, for neutron star surfaces approximately 10^{-10}, and near Sgr A* of order 10^{-5}. These deviations are in principle detectable with next-generation atomic clocks and very-long-baseline interferometry.

7.4 Gravitational-Wave Echoes

After a binary merger, the remnant torsion star possesses a stratified interior with alternating attractive and repulsive layers. Gravitational waves partially reflect at each torsional interface, producing a train of echoes. The time delay between successive echoes is

$$\Delta t \sim \frac{GM}{c^3} \approx 10^{-5}\left(\frac{M}{10M_\odot}\right)\ \text{s}.$$

The echoes carry a characteristic phase modulation determined by the dual-rotor spectrum. Detection of such echoes by LISA or the Einstein Telescope would constitute direct evidence for the discrete torsional structure of strong gravity.

7.5 Cosmological Implications

The early universe is described as a single primordial torsion star. The Planck-era torsional mismatch drives a brief de Sitter-like expansion without requiring an inflaton field. Reheating occurs through layer transitions that convert torsional energy into radiation. The resulting primordial power spectrum is naturally scale-invariant, arising from the discrete rotor modes rather than from quantum fluctuations of a scalar field. Dark matter is unnecessary: galactic rotation curves are explained by residual baryonic torsional misalignment at galactic scales. Dark energy emerges as the minimum non-zero vacuum torsional rigidity $J_{\text{vac}} > 0$ induced by quantum rotor fluctuations, yielding a cosmological constant of order $\Lambda \sim \ell_P^{-2}$.

All of the above predictions are falsifiable within the next decade. Non-observation of strong-field deviations at the predicted level, absence of gravitational-wave echoes, or detection of a true event horizon would rule out the framework. Conversely, positive detection of any of these signatures would confirm that spacetime is discrete, particle-local, and doubly temporal at the most fundamental level.

8 Comparison with Other Quantum Gravity Approaches

The double spacetime theory occupies a distinctive position among quantum gravity candidates. While most frameworks attempt to quantize or discretize an a priori continuum geometry, the present approach derives both quantum mechanics and gravity from the intrinsic 16-dimensional biquaternion algebra carried by every massive particle. Below we compare the theory with the principal existing approaches, highlighting the conceptual and technical differences.

8.1 Loop Quantum Gravity

Loop quantum gravity (LQG) discretizes spacetime via spin networks and quantizes area and volume operators while retaining a background manifold and Riemannian curvature as fundamental. Singularities are resolved by a minimal eigenvalue of the area operator, yet the continuum limit remains challenging and the dynamics are encoded in complicated Hamiltonian constraints.

In contrast, the double spacetime theory imposes discreteness directly on the dual-rotor angles $\phi_a, \theta_a \in (2\pi/N)\mathbb{Z}$, without ever introducing a background manifold. Curvature is absent; gravity is torsional mismatch in the Lie algebra $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$. The reconstruction problem of LQG—recovering a smooth spacetime from spin networks—does not arise, because macroscopic spacetime emerges collectively from the torsional alignment of neighboring particle-intrinsic dual rotors.

8.2 String Theory

String theory unifies forces by embedding gravity in a ten- or eleven-dimensional spacetime populated by vibrating strings and requiring supersymmetry. While mathematically elegant, it suffers from a vast landscape of vacua and lacks unique low-energy predictions.

The double spacetime theory achieves complete unification within fixed four-dimensional $\mathrm{Cl}(3,1)$ without extra dimensions or supersymmetry. Gravity (usual-dual boost mismatch), electromagnetism (dual rotation gauge coupling), the weak force (chiral dual gauge), and the strong force (layered torsional dynamics) all emerge as different modes of the same rotor algebra. No Calabi–Yau compactification or brane constructions are required; the dual spacetime is intrinsically compactified at the particle level.

8.3 Canonical and Ashtekar Approaches

The Ashtekar reformulation casts general relativity in terms of self-dual connections, enabling a Yang–Mills-like quantization. However, it inherits the continuum singularities of classical gravity and must confront the Wheeler–DeWitt equation with its notorious factor-ordering and regularization problems.

In the present framework the connection is eliminated entirely. Parallel transport is replaced by rotor exponentiation, and the Killing form on the torsional bivector replaces the curvature scalar. The resulting theory is constraint-free at the classical level and quantizes naturally via the finite rotor group, avoiding the Wheeler–DeWitt equation altogether.

8.4 Emergent Gravity and Causal Sets

Emergent gravity approaches treat spacetime as a thermodynamic or entropic phenomenon, while causal set theory discretizes Lorentzian order. Both frameworks struggle to recover the exact Einstein equations and face fermion doubling or chirality issues.

The double spacetime theory recovers the Einstein equations precisely through its equivalence with teleparallel gravity, while fermions arise naturally as excitations of the biquaternion minimal left ideal. Discreteness is algebraic (Diophantine constraints on rotor angles) rather than set-theoretic, and causality is preserved by the lightlike $J = 0$ resonance condition. No thermodynamic analogy is invoked; the emergence of spacetime is a collective effect of torsional alignment among particle-intrinsic dual rotors.

In summary, existing quantum gravity programs extend or quantize the continuum manifold while retaining curvature as fundamental. The double spacetime theory abandons both the manifold and curvature, deriving spacetime, gravity, and quantum states from the single algebraic structure of dual rotors in $\mathrm{Cl}(3,1)$. It thereby resolves singularities, unifies all interactions, and renders gravity controllable without introducing new dimensions, supersymmetry, or additional fields.

9 Experimental Testability from the Quantum Perspective

Although the double spacetime theory is fundamentally algebraic, its quantum sector makes concrete experimental predictions that can be tested with current or near-future technology. The key insight is that the torsional mismatch scalar J depends solely on the relative angle between the usual and dual rotors. External fields capable of coherently modulating the dual-rotor phases ϕ_a can therefore alter gravitational and inertial effects in a measurable way.

9.1 Theoretical Coupling Mechanism

Circularly polarized electromagnetic fields in the THz range (10^{13}–10^{15} Hz) possess helical wavefronts that couple preferentially to the dual generators $\Gamma_a = I, J, K$. Because the dual spacetime is time-reversed, the helicity $\sigma = \pm 1$ of the wave directly controls the direction of dual-rotor drive: right-handed polarization reinforces attractive torsional mismatch ($J > 0$), while left-handed polarization opposes it. The effective interaction Hamiltonian is

$$H_{\mathrm{int}} = \kappa \sigma E_0^2 \sum_a \phi_a \Gamma_a,$$

where E_0 is the electric-field amplitude and κ is a material-dependent coupling constant. Near resonance, the dual phase evolves according to

$$\dot{\phi} = \kappa \sigma E_0^2 \sin(\omega t - \phi + \delta_\sigma).$$

9.2 Proposed Near-Term Experiments

The most accessible platform exploits collective coherence in superconductors or Bose–Einstein condensates, where macroscopic numbers of particles ($N \sim 10^{26}$) can lock their dual rotors into a common phase. A concrete proposal is as follows:

- Suspend a microgram-scale superconducting test mass inside an asymmetric THz resonator (quantum cascade laser or free-electron laser cavity) with precise circular-polarization control.
- Monitor the apparent weight or inertial mass via a high-precision torsion balance or superconducting gravimeter while sweeping the frequency and helicity.
- Look for resonant, helicity-dependent changes in effective mass at the predicted dual-rotor resonance frequencies $\theta \sim c/(2\pi r_{\mathrm{comp}})$, where r_{comp} is the compactification radius set by the particle mass.

Predicted effect sizes are $\Delta m/m \sim 10^{-2}$–10^{-5} at achievable intensities of 10^{12}–10^{15} W/m^2. The signature is distinctive: the effect must reverse sign when the polarization helicity is flipped and must vanish for linearly polarized or unpolarized radiation.

9.3 Falsifiability and Caveats

A null result at the 10^{-6} level after a systematic frequency and helicity scan would constrain the dual-rotor coupling strength and the compactification scale. Positive detection—particularly a helicity-reversible, resonance-peaked mass shift—would constitute strong evidence that gravity and inertia are torsional phenomena originating in dual spacetime.

The theory makes no claim of immediate technological breakthrough. The proposals above are modest, table-top experiments whose primary purpose is to test the fundamental hypothesis that dual-spacetime degrees of freedom couple measurably to laboratory electromagnetic fields. Success at this level would open the door to larger-scale gravitational engineering; failure would simply bound the parameters of the framework.

9.4 Broader Implications

If dual-rotor control proves feasible, the same mechanism offers pathways to nuclear transmutation and radioactivity neutralization by driving torsional layer transitions inside atomic nuclei. These applications, however, lie beyond the scope of the present quantum-gravity test program and will be addressed in future work.

In summary, the double spacetime theory transforms gravity from an immutable geometric constraint into a controllable torsional degree of freedom whose quantum sector is already within experimental reach. The coming decade will decide whether nature has granted us access to this new degree of freedom.

10 Conclusions

The double spacetime theory presented in this work offers a complete algebraic framework for quantum gravity. By recognizing that every massive particle carries an intrinsic pair of compactified Minkowski spacetimes encoded in the 16-dimensional biquaternion algebra $\mathbb{H} \oplus \mathbb{H} \cong \mathrm{Cl}(3, 1)$, we have derived both quantum mechanics and gravity from a single geometric structure without invoking a background manifold, Christoffel symbols, or Riemann curvature.

Quantum states have been shown to be rotations in the compact dual spacetime: the dual rotor R_{dual} is mathematically identical to the SU(2) spin-1/2 rotation operator, and the Hilbert space emerges directly from the space of dual-rotor phases. The Dirac equation arises covariantly from the action of the complete rotor group on the biquaternion minimal left ideal, with fermion masses originating from torsional rigidity between the usual and dual rotors. The measurement problem and entanglement receive purely geometric interpretations as torsional synchronization and correlated dual phases, respectively.

Gravity itself is reinterpreted as the dynamical cost of misalignment between the usual and dual rotors. The Lorentz-invariant scalar J constructed from the Killing form on the torsional mismatch bivector yields an action dynamically equivalent to the Einstein–Hilbert action, yet the underlying ontology is entirely algebraic. Singularities are forbidden by the boundedness of discrete rotor angles, and event horizons do not form because light rays can tunnel through finite torsional layers.

In this framework, general relativity is not superseded but completed. Its empirical successes are preserved exactly in the classical limit, while its conceptual difficulties—singularities, the origin of inertia, the measurement problem, and the nature of spacetime itself—are resolved at the algebraic foundation. Quantum mechanics and gravity are revealed as two aspects of the same 16-dimensional structure: quantum mechanics is the kinematics of dual-spacetime rotations, and gravity is the dynamics of their relative misalignment.

The continuum was a brilliant classical approximation. Reality, at the deepest level, is discrete, particle-local, and doubly temporal. Spacetime does not curve matter; matter misaligns its own intrinsic dual spacetimes, and this misalignment, collectively perceived, is what we have called gravity.

With this reformulation, the century-long quest for quantum gravity reaches its geometric conclusion within the 16 dimensions of the biquaternion algebra.

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